

Q1

```
main.cpp  Run  Output  Clear
29 int main(){
30     int arr[] = {2, 5, 8, 12, 16, 23, 38, 56, 72, 91};
31     int size = sizeof(arr)/sizeof(arr[0]);
32     Solution s1;
33     int ans = s1.binarySearch(arr,size,23);
34     if(ans != -1){
35         cout<<"23 found at index : "<<ans<<endl;
36     }
37     else{
38         cout<<"23 is not there in the array ! "<<endl;
39     }
40     return 0;
41 }
```

23 found at index : 5

=== Code Execution Successful ===

Q2

```
main.cpp  Run  Output  Clear
62 int main(){
63     int arr1[] = {12, 11, 13, 5, 6, 7};
64     int size1 = sizeof(arr1)/sizeof(arr1[0]);
65
66     int arr2[] = {38, 27, 43, 3, 9, 82, 10};
67     int size2 = sizeof(arr2)/sizeof(arr2[0]);
68
69     Solution s1,s2;
70     s1.mergesort(arr1,0,size1-1);
71     s2.mergesort(arr2,0,size2-1);
72
73     cout<<"After applying merge sort on first array : "<<endl;
74
75     cout<<"After applying merge sort on second array : "<<endl;
76 }
```

After applying merge sort on first array :
7 11 11 12 12 13

After applying merge sort on second array :
3 3 10 27 27 27 38

=== Code Execution Successful ===

Q3

```
main.cpp  Run  Output  Clear
34 int main(){
35
36     int arr[] = { 4, 2, 6, 9, 2 };
37     int size = sizeof(arr)/sizeof(arr[0]);
38     Solution s1;
39     s1.quicksort(arr,0,size-1);
40     cout<<"Sorted array by using quick sort algo : ";
41     for(int i=0 ; i < size; i++){
42         cout<<arr[i]<<" ";
43     }
44     cout<<endl;
45     return 0;
46 }
47
```

Sorted array by using quick sort algo : 2 2 4 6 9

=== Code Execution Successful ===

Q4

```
main.cpp  Run  Output  Clear
20 };
21 int main(){
22     int arr[] = {-2,-5,6,-2,-3,1,5,-6};
23     int size = sizeof(arr)/sizeof(arr[0]);
24
25     Solution s;
26     int ans = s.max_sub_array_sum(arr,size);
27     cout<<"maximum subarray sum is : "<<ans<<endl;
28     return 0;
29 }
30
31
32
```

maximum subarray sum is : 7

=== Code Execution Successful ===